

Atherosclerosis and oxidant stress: the end of the road for antioxidant vitamin treatment?

Extensive experimental data have revealed a central role for oxidative stress in atherogenesis and suggested a potential role for 'antioxidant' treatment in cardiovascular disease (CVD) [1-11]. Experimental data, however, have not translated into clinical benefit: most antioxidant vitamin trials have failed to reduce cardiovascular morbidity and mortality [12]. Moreover, recent clinical trials have suggested that mono-therapy with certain antioxidant vitamins like vitamin E may, in fact, be detrimental [13]. As a result of the disappointing outcome of 'antioxidant' vitamin trials, some authors have questioned both the utility of 'antioxidant' treatment in CVD and the supposedly central role of oxidative stress in atherogenesis [14-19]. Other investigators, however, sustain that the beneficial effects of lipid lowering and anti-hypertensive treatment are at least, in part, due to their 'antioxidant' properties, in addition to their specific pharmacological properties [20, 21]. Oxidant stress plays a pivotal role in atherogenesis, however, the clinical promise of antioxidant vitamins has failed to translate into clinical benefit. Increasing evidence suggests that more rigorous clinical trial designs are necessary to effectively divulge antioxidant utility and that a multifaceted antioxidant approach to atherosclerosis may yield the most clinical reward. This article reviews currently available evidence on the role of oxidant stress in atherosclerosis, analyzes the results of large anti-oxidant trials, and suggests ways to investigate the true role of antioxidant treatment in the clinical setting.